



Automate Deployment with GitHub Actions

Use GitHub Actions to:

- Test your code
- Build Docker images
- Deploy automatically

Step 1: Create Workflow File

Create `.github/workflows/deploy.yml`

```
name: Deploy Flask App

on:
  push:
    branches: [ main ]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v3
      - name: Set up Python
        uses: actions/setup-python@v4
        with:
          python-version: '3.12'
      - name: Install dependencies
        run: pip install -r requirements.txt
      - name: Run tests
```



Add Docker Build + Deploy

Extend the workflow:

- name: Build Docker image
run: docker build -t my-flask-app .
- name: Deploy to Server
uses: appleboy/ssh-action@v1
with:
 - host: \${ secrets.HOST }
 - username: \${ secrets.USER }
 - key: \${ secrets.SSH_KEY }
 - script: |
 - cd ~/my-flask-app
 - git pull origin main
 - docker build -t flask-app .
 - docker stop flask-running || true
 - docker rm flask-running || true
 - docker run -d --name flask-running -p 80:5000 flask-app

Store Secrets in GitHub

Go to your repo → Settings → Secrets → Actions

Add:

- `HOST` : your server IP/domain
- `USER` : SSH username
- `SSH_KEY` : private key for server access

 **CI/CD Ready with Every Push!**